

Yash Kherwal

+91 9891148156 | yash.kherwal78@gmail.com | linkedin.com/in/yash-kherwal-944497254

EDUCATION

Indian Institute of Technology Roorkee

2022 – 2026

B.Tech in Chemical Engineering

EXPERIENCE

Software Development Intern

May 2025 – June 2025

ScoreMe Solutions

- Designed a **two-stage PDF classification pipeline** to categorize fintech documents as scanned, digital, or hybrid — a critical preprocessing step for **structured data extraction** workflows.
- Built a **rule-based heuristic engine** using **PDFBox** computing **image-area ratios**, **word counts**, and **OCR confidence** via **Tesseract** to produce high-confidence labels, flagging ambiguous documents for **ML classification**.
- Trained a **Random Forest classifier** on features including **text density**, **image presence**, **font metadata**, and **page structure** to accurately classify edge cases such as mixed-page and ad-heavy PDFs.
- Extending pipeline with a **confidence threshold** layer — documents below 70% model confidence flagged for manual review, improving reliability of downstream data extraction at scale.

Engineering Intern

June 2025 – Jul 2025

Bharat Electronics Limited

- Developed a **real-time packet analyzer** for **ASTERIX CAT048** radar protocol, processing live surveillance streams for aviation data monitoring.
- Implemented **binary parsing** and **protocol decoding** to convert raw radar packets into structured aviation records, enabling downstream analysis.
- Optimized **low-level stream processing** to improve system reliability and reduce packet loss under high-throughput conditions.

PROJECTS

Semantic Document Search System (GDSC IIT Roorkee) | *Python, LangChain, Sentence-Transformers, AstraDB*

- Identified that **keyword-based search** failed on GDSC's 500+ PDF knowledge base due to vocabulary mismatch — built a **semantic retrieval system** to solve this.
- Used **LangChain document loaders** and **RecursiveCharacterTextSplitter** for chunking; generated **dense vector embeddings** via **Sentence-Transformers** stored in **AstraDB** for **approximate nearest-neighbor search**.
- Built **RetrievalQA** chain connecting **AstraDB vector store** to an **LLM** with **custom prompt templates**, enabling grounded answer synthesis and reducing hallucinations on multi-document queries.
- Achieved measurable improvement in **answer relevance** on club-specific queries where **BM25** baseline returned zero useful results.

SC-MFC Power & Cost Optimisation | *Python, TensorFlow/Keras, Scikit-learn, pdfplumber*

2025 – 2026

- Addressed the challenge of nonlinear, multi-variable power optimisation in **Single-Chamber Microbial Fuel Cells** where traditional one-factor-at-a-time methods miss interaction effects.
- Extracted **real experimental data** from research PDFs using **pdfplumber**; applied **median imputation**, **sanity bounds filtering**, and **StandardScaler** normalization to handle missing values and prevent data leakage.
- Trained a **multi-output ANN** with **BatchNorm** and **Dropout** layers using **Adam optimizer**, **EarlyStopping**, and **ReduceLROnPlateau** to jointly predict **power density** and **cost index** from electrochemical input features.
- Performed **Pareto-style random search** over candidate configurations scored by a **power-to-cost ratio**, identifying the optimal combination of pH, temperature, electrode spacing, and substrate concentration.

R-Land Hunt | *Node.js, Express, EJS, Geolocation API* | *MDG Space, IIT Roorkee*

Dec 2023 – Jan 2024

- Solved the problem of physical QR-code setup and on-site chaos at IIT Roorkee scavenger hunts by building a **location-based web application** that unlocks clues only when a participant is physically present at the correct location.
- Used the browser's **Geolocation API** to capture participant GPS coordinates and validated them against stored clue coordinates server-side in **Express**, unlocking clues only on coordinate match.
- Designed **role-based interfaces** in **EJS** — admins configure clue locations and manage scores pre-event; participants collect clues and spend **negative-point hints** with a **real-time leaderboard** reflecting scores instantly.

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL

ML / AI: TensorFlow, Scikit-learn, Sentence-Transformers, LangChain, NLTK

Libraries: Pandas, NumPy, OpenCV, pdfplumber

Tools & Platforms: Git, FastAPI, Streamlit, AstraDB, Tesseract OCR, PDFBox, Node.js, Express